

AI-Driven Two-Tier Landslide Early Warning & Disaster Resilience Grid for North-East India (NER)

XGBoost Two-Tier Stacking • Meta-Calibrator Fusion • Real-Time IoT Telemetry • InSAR Ground Deformation • Dijkstra Routing • OASIS CAP v1.2

Dr. R. Rajmohan | Disaster Management Research Group, SRM Institute of Science & Technology | 20°30'N, 88°-98°E | NH-10, NH-27, NH-29, NH-44

Artificial Intelligence

Geotechnical Eng.

Satellite Remote Sensing

Disaster Risk Reduction

1 Study Context

North-East India records >200 fatalities per monsoon season from landslides across 8 national highways. Existing rainfall-threshold systems lack spatial specificity and probabilistic calibration.

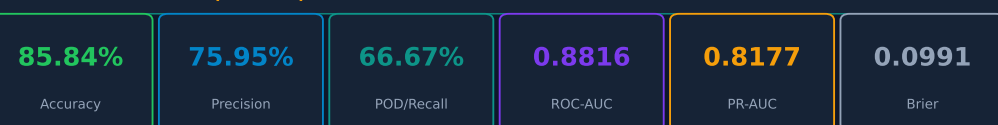
MINDMELD decouples static terrain susceptibility $S(x)$ from dynamic hydro-geotechnical triggers $T(x,t)$ using a novel two-tier XGBoost stacking architecture fused by a meta-calibrator, yielding calibrated $P_{fused} \in [0,1]$ mapped to 4 NDMA operational alert tiers.

4 Performance Metrics Dashboard

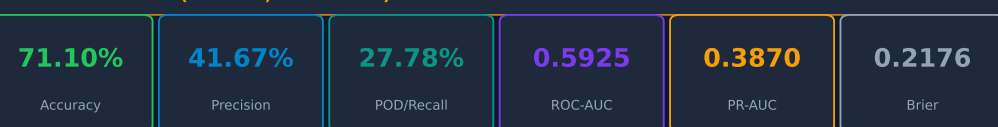
TEST PARTITION (N = 52)



SPATIAL BLOCK OOF (N = 346)



LOYO TEMPORAL (N = 346, 2007-2019)



5 Confusion Matrix — Test Partition ($\theta = 0.35$)

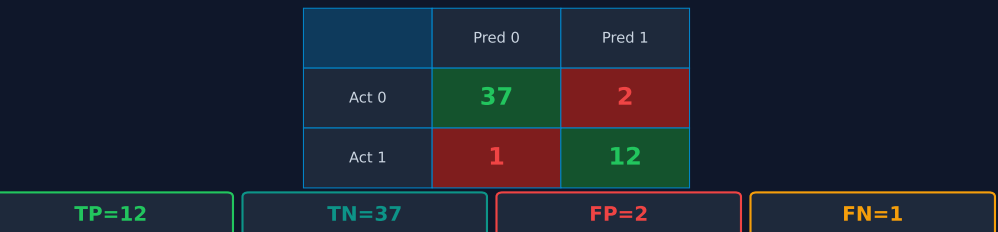


Fig 1 · Confusion Matrix Heatmap

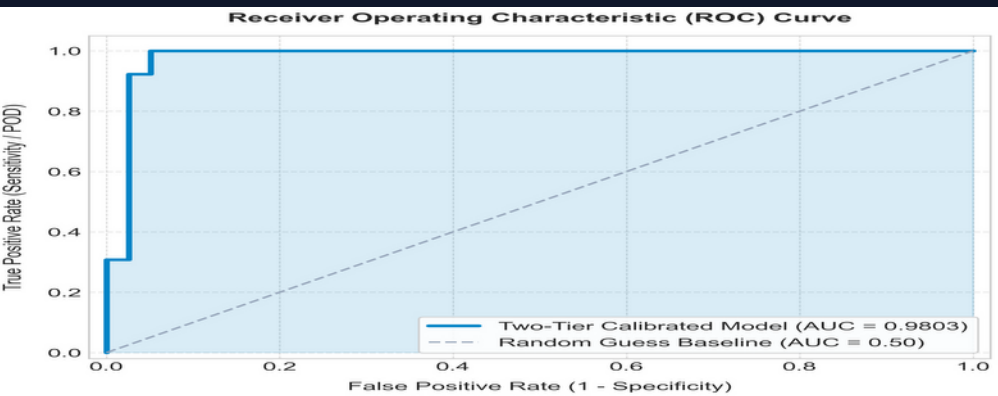
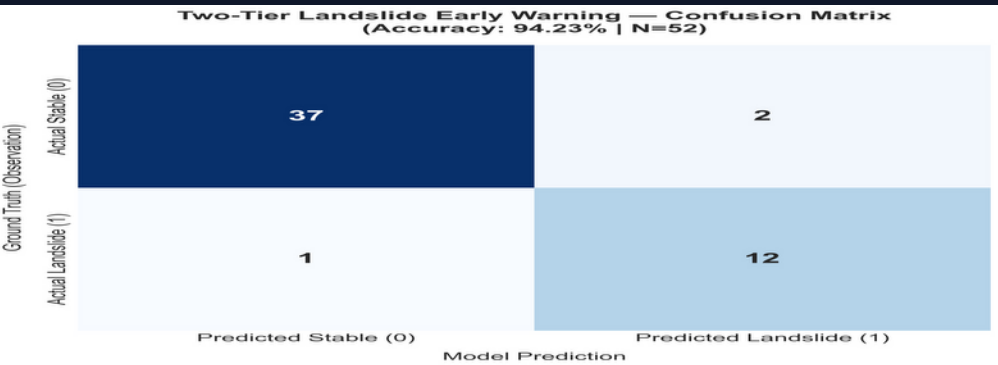


Fig 2 · ROC Curve | AUC = 0.9803

6 Global TreeSHAP Feature Attribution (Trigger Model)

rain_48h_prior 1.8878 (34.7%)

r24_seasonal_anom 0.8781 (16.1%)

rain_24h_obs 0.7105 (13.1%)

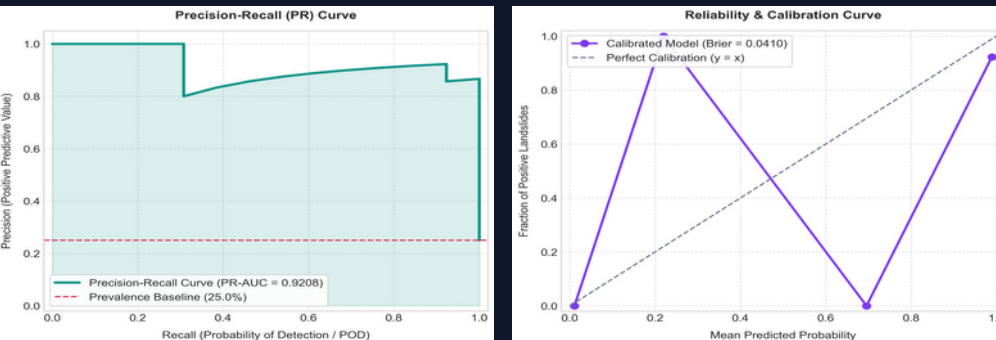
rain_7d_prior 0.6528 (12.0%)

api_seasonal_anom 0.5773 (10.6%)

rain_72h_prior 0.4913 (9.0%)

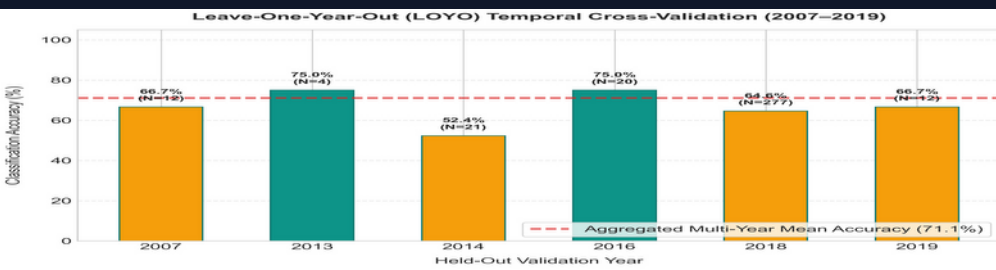
api_7d 0.2419 (4.5%)

7 Precision-Recall & Calibration Curves



8 Leave-One-Year-Out Temporal Validation (2007-2019)

Year	N	Acc%	Prec%	POD%	FAR%	AUC
2007	12	66.67	0.00	0.00	100.0	0.222
2013	4	75.00	0.00	0.00	0.00	1.000
2014	21	52.38	0.00	0.00	100.0	0.378
2016	20	75.00	0.00	0.00	0.00	0.400
2018	277	64.62	23.53	16.90	76.47	0.564
2019	12	66.67	33.33	33.33	66.67	0.296
MEAN	346	71.10	41.67	27.78	58.33	0.593



Dijkstra Ev-Router
Interferogram unwrapping
Real-Time System Components
Sentinel-1A InSAR

CAP v1.2 Alert Engine
Piezometers + MEMS IPI
40-Node IoT Grid

3 Geotechnical Physics — Infinite Slope FS

Terzaghi Effective Stress

$$\sigma' = (\gamma_s z - \gamma_w h_w) \cdot \cos^2 \beta$$

Mohr-Coulomb Shear Strength

$$\tau_f = c' + \sigma' \cdot \tan \phi'$$

Infinite Slope FS

$$FS = [c' + (\gamma_s z - \gamma_w h_w) \cos^2 \beta \cdot \tan \phi'] / [\gamma_s z \cdot \sin \beta \cdot \cos \beta]$$

Critical Rainfall h_w^*

$$h_w^* = z - [c' / (\gamma_w \cos^2 \beta \cdot \tan \phi')] + [z \tan \beta / \tan \phi']$$

Richards PDE

$$\partial \theta / \partial t = \partial / \partial z [K(h) (\partial h / \partial z + 1)] - S(z, t)$$

API Decay

$$API_t = \sum \{P_{t-k}\} \cdot \exp(-k/\tau) \quad [\tau = 7 \text{ days}]$$

Seasonal Anomaly

$$Anom_t = (P_t - \mu_{clim}(doy)) / \sigma_{clim}(doy)$$

10 TreeSHAP Attribution & Risk Mapping

